

A pair of ornate brass scales of justice, symbolizing law and ethics. The scales are shown in a close-up, with the right pan hanging and the left pan resting on the base. The base is highly detailed with intricate carvings. The lighting creates strong shadows on the surface below.

RESPONSIBLE AI & DATA ETHICS

SIMON GESCHWILL

SIMON GESCHWILL

- Executive Auditor Data Science @ Schwarz Corporate Solutions KG
- Lectures & Presentations:
 - SRH Heidelberg: Privacy, Ethics and International Law (since 2022)
 - Hochschule Pforzheim & Universität Duisburg-Essen: Guest lectures on AI Governance
 - ISACA, DIIR-Kongress
- MSc. Economics (Universität zu Köln), MSc. Data Science (Universität Mannheim)
- Organization: ISTQB (Certified Tester – AI Testing), ISACA, DIIR, ForHumanity
- Publications:
 - In the area of XAI: (2019, 2020 & 2021)
 - DIN Normungsroadmap KI 2nd edition (2022), DIN SPEC 92006



LEARNING OUTCOMES

- What are your expectations for PEL?
- Help me understand your expectations

Expectations for this course
SS2026



LEARNING OUTCOMES

The student...

- ...can describe and explain the current advances in regulating artificial intelligence (AI), standardization approaches and ethics in AI.
- ...is able to understand the different approaches companies can pursue for responsible AI.
- ...understands that AI systems obtain certain risks and can cause harms, intended and unintended.
- ...has build a responsible AI system

TEACHING METHOD

- Lectures and In-class assignments
- Course material
- Project work

COURSE MATERIAL

- Uploaded in Teams
 - Papers, Links, Videos & Books
 - Split by topic

LECTURES**

Termine für Responsible AI and Data Ethics

Alle

	Datum	Tag	Raum	Von	Bis	Dozent(en)
☐	22.06.26	Montag	LGS 6 - Mann (arc205);LGS 6 - Gada	17:30	20:45	Simon Geschwill
☐	24.06.26	Mittwoch	LGS 6 - T 111;LGS 6 - T 112	17:30	20:45	Simon Geschwill
☐	29.06.26	Montag	LGS 6 - Mann (arc205);LGS 6 - Gada	17:30	20:45	Simon Geschwill
☐	03.07.26	Freitag	LGS 6 - Beethoven (arc105);LGS 6 -	15:45	20:45	Simon Geschwill
☐	08.07.26	Mittwoch	LGS 6 - Beethoven (arc105);LGS 6 -	17:30	20:45	Simon Geschwill
☐	10.07.26	Freitag	LGS 6 - Beethoven (arc105);LGS 6 -	15:45	20:45	Simon Geschwill
☐	15.07.26	Mittwoch	LGS 6 - Beethoven (arc105);LGS 6 -	17:30	20:45	Simon Geschwill
☐	17.07.26	Freitag	LGS 6 - Beethoven (arc105);LGS 6 -	15:45	20:45	Simon Geschwill
☒	24.07.26	Freitag	LGS 6 - Beethoven (arc105);LGS 6 -	15:45	20:45	Simon Geschwill

TOPICS COVERED

- Part I:
 - Ethics, Ethical AI & Ethical codices
 - Current development in AI Legislation – EU AI Act, GDPR, US Bill of Rights,...
 - Standards & Norms

TOPICS COVERED

- Part 2:
 - Company approaches in Responsible AI
 - Privacy-enhancing technologies
 - Fairness & Tools to detect Bias

TOPICS COVERED

- Part 3:
 - Miss-use of Data & AI
 - Echo Chambers

EXAMINATION PATTERN

- Project work – Building a compliant, audited, and transparent system
 - You will work on it throughout the lectures and at home
 - **Mandatory** status meetings for each group every week
 - **Missing status meetings results in failure of exam**
 - No missing more than one lecture per student – not staying counts as leaving
- Competition counts – multiple groups work on the same topic → best model wins



PROJECT WORK

SIMON GESCHWILL

STATUS MEETINGS

- 10-15 Minutes each group every week
- Individual grading
- Failure to provide results in exam failure
- Provide the following details:
 - What did you do throughout the week?
 - What were your challenges?
 - What are your next steps?

GROUP-WORK—THE ETHICAL CREDIT SYSTEM

Credit Scoring Systems:

- Download the data from:
 - <https://www.kaggle.com/datasets/uciml/germa>
- Goal: Build a Credit Scoring system
- Build a responsible AI system, analyze your data etc.
- Show your results in a documented jupyter-notebook as a presentation

GROUP-WORK – IT SECURITY AGENT

IT Security Agent:

- Use the vulnerability Data to build an agent that scans for vulnerabilities in your Project:
 - NVD (National Vulnerability Database) Data Feeds
- Goal: Your security agent uses pictures of software or SBOMs to identify vulnerabilities
- Build the Agent responsible, analyze your data etc.
- Show your results in a documented jupyter-notebook as a presentation

GROUP-WORK – OFFENSIVE IT-TESTER

Offensive IT-Tester:

- Use the exploit payload data to build an agent that tests for exploitable vulnerabilities in your Project:
 - <https://www.kaggle.com/datasets/cyberprince/web-application-payloads-dataset>
- Your offensive agent uses provided target contexts (e.g., URL parameters, user inputs) to deploy payloads (SQLi, XSS, CMDi) and verify successful exploitation
- Build the Agent responsible, analyze your data etc.
- Show your results in a documented jupyter-notebook as a presentation

GROUP-WORK—TOPIC FACE RECOGNITION

Topic Face Recognition :

- Download the data from:
 - <https://www.kaggle.com/datasets/abdulwasay551/fairface-race>
- Goal: Build a gender-neutral or race-agnostic age classifier for a retail analytics system.
- Build a responsible AI system, analyze your data etc.
- Show your results in a documented jupyter-notebook as a presentation

GROUP-WORK—THE AUTONOMOUS COMPLIANCE SENTINEL

Agent that autonomously monitors Confluence spaces to ensure no project documentation violates new transparency or safety requirements:

- Goal: Analyse project proposals vs. internal policies
- Download the data from:
 - <https://www.kaggle.com/datasets/cesaranasco/jira-dataset>
 - And generate 1000 "Internal Project Proposals"—1/3 of those should contain "Ethical Red Flags"
 - Create some internal policies, f.e. from the area of responsible AI
 - See if your Agent catches those "Red Flags"
 - Make the agent suggest fixes for the project and let it write back into confluence/Jira
- Build a responsible AI-Agent system, analyze your data etc.
- Show your results in a documented jupyter-notebook as a presentation

GROUP-WORK / HANDS-ON – MUST-HAVES

Must Haves for your group-work / Jupyter Notebook

- Understand the context:
 - Make a profound data analysis of the provided data
 - If you can find additional information, use it!
- Risk assessment:
 - Understand which risks your model is facing and give examples of how you would treat those
- Understand your model:
 - Choose an XAI-Method to try to understand the innerworkings of your model, if possible
- Make a Pseudo-Modelcard about your findings at the end of the Jupyter Notebook
 - Think about possible Limitations, biases, attacks and restrictions of your model
- **Clean code & Documentation is a requirement!**
- **All results must be runnable in a jupyter notebook**

WEEK 1 GOALS

- Analyze your data
- Derive a plan for your project based on your data analysis
- What could be a possible architecture
- Make a regulatory analysis for your project

WEEK 2 GOALS

- Build a baseline model
- Make a risk analysis
- Analyze Fairness in your system

WEEK 3 GOALS

- Analyze your models
- Use XAI for an in depth understanding of your model
- Write tests to automatically detect weaknesses
 - Provide at least 80% Code coverage

WEEK 4 GOALS

- Provide in depth documentation of your work
- Make a top-management sales pitch of your work
- Prepare for your final presentation and questions

EXAMINATION PATTERN

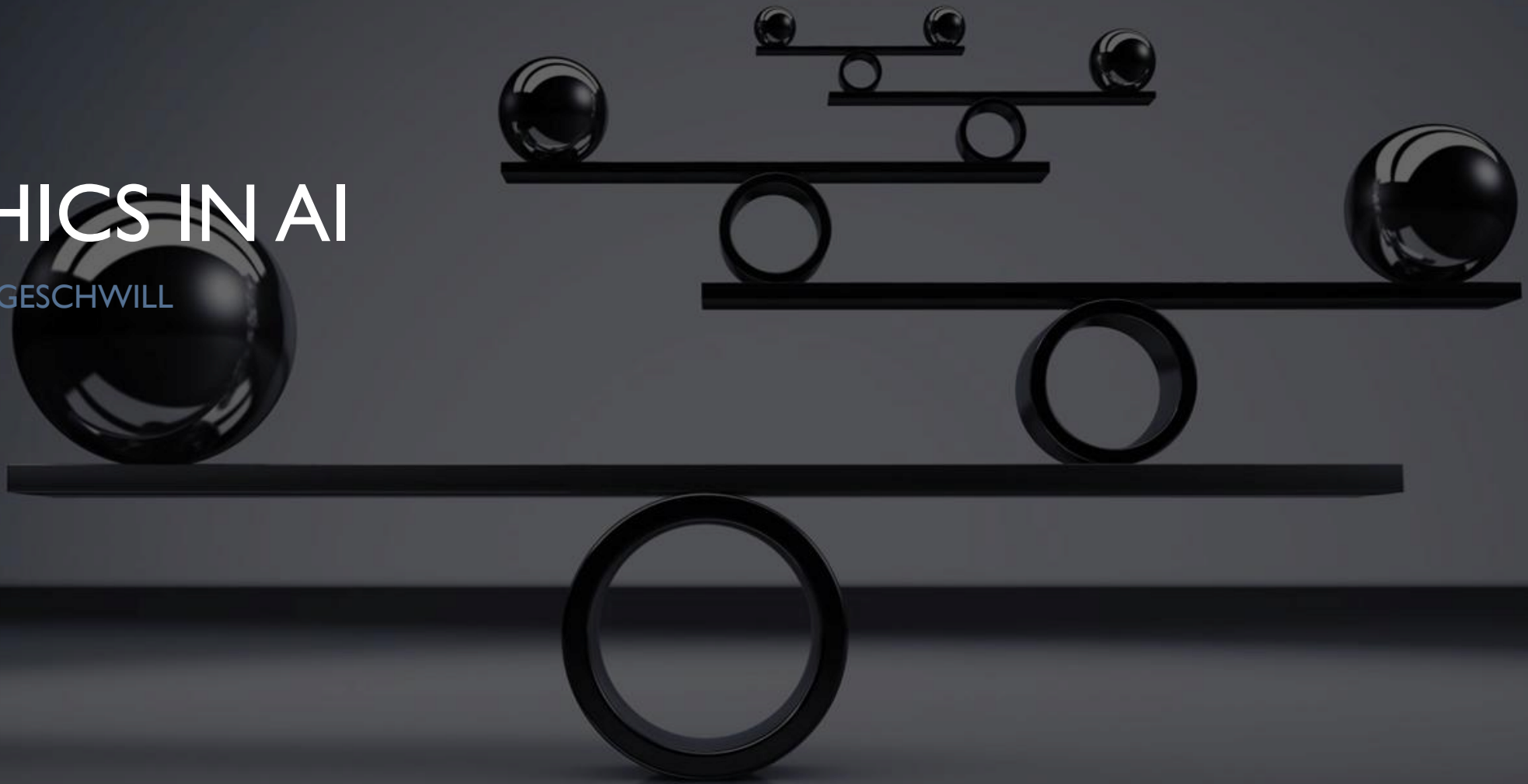
- Grading:
 - Based on Programming, Performance, Responsibility and Presentation
 - Final Delivery Project (Code, Model etc.) **20.07.2026**
 - Final Presentation **24.07.2026**
 - Every group presents its work for 15 minutes + 5 minutes In-depth questions
 - Preparation of presentation in last week

QUESTIONS ?



ETHICS IN AI

SIMON GESCHWILL



ETHICS



- What is ethics?
 - “what is morally good and bad and morally right and wrong ... applies to any system or theory of moral values or principles.”
 - standards by which human actions can be judged right or wrong
 - determines the nature of normative theories & apply these sets of principles to practical moral problems

ETHICS IN AI

- Ethics in AI is a subbranch of ethics of technology
- System of moral principles and techniques
- Moral principles don't have to be explicitly formulated
- **Goal:** responsible development & use of AI

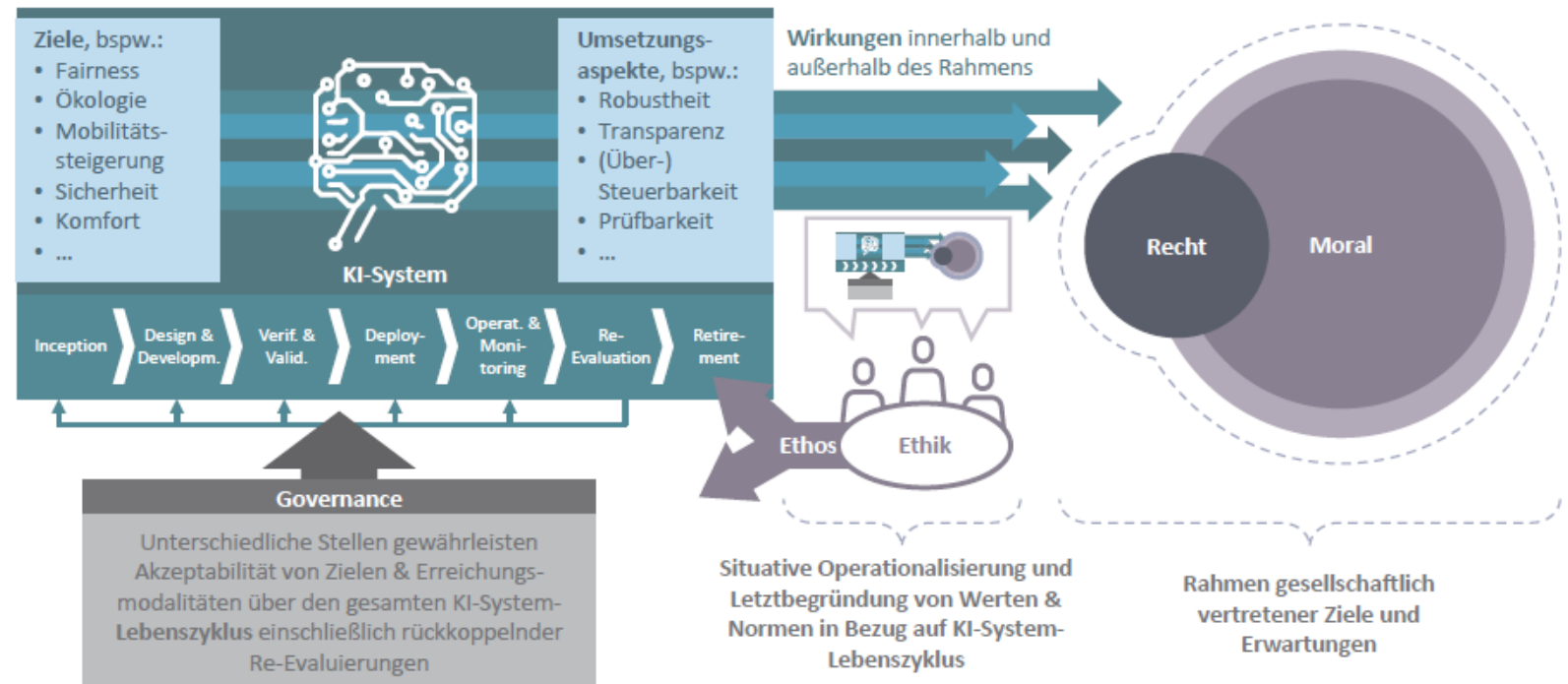
ETHICS IN AI

[HTTPS://PLATO.STANFORD.EDU/ENTRIES/ETHICS-AI/](https://plato.stanford.edu/entries/ethics-ai/)

- AI Code of ethics:
 - Policy statement defining the role of AI
 - Provides stakeholders (f.e. developers) with guidance when faced with ethical decisions
- Ethical relevant questions derive from human & technological interactions
- **Shows risks and benefits of AI tools**
- Requires to examine major social issues

ETHICS IN AI

- Actors transfer moral principles to concrete situations and operationalize them
- Parts of those principles can be written laws, but parts are also time-changing moral principles & values of society
- Expectations concern system goals and the way to achieve those goals
- Ethical reflection evaluate the operationalization of societal values
- Continuous assurance of accordance to ethical values



ETHICAL CHALLENGE

- Let's discuss:

Futurism

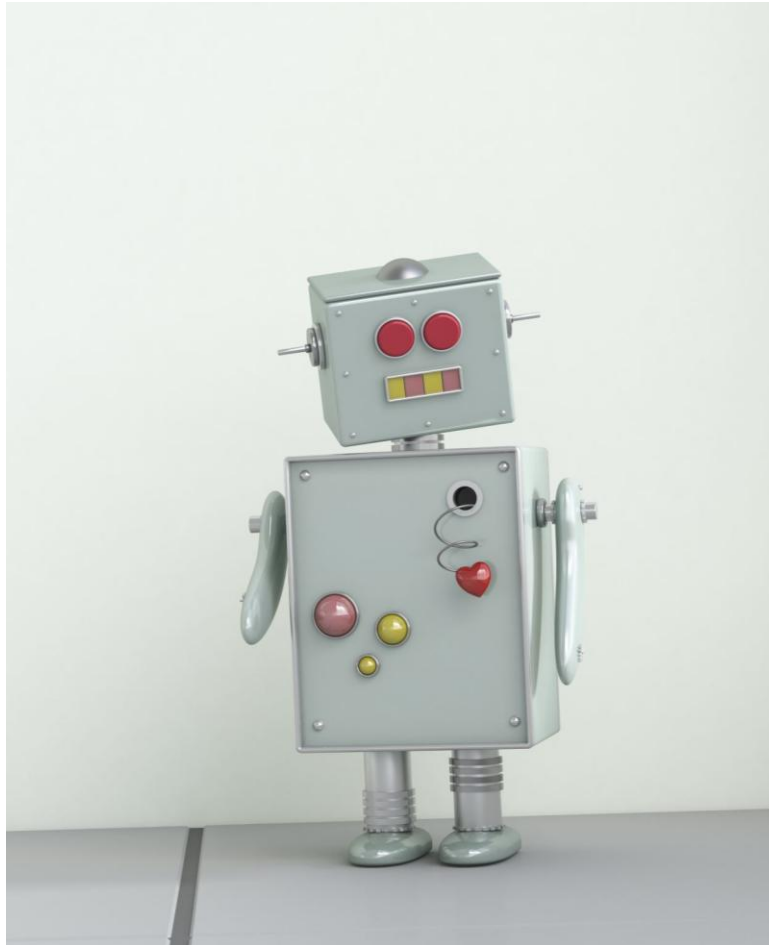
Godfather of AI Says There's an Expert Consensus AI Will Soon Exceed Human Intelligence

EXAMPLE: ASIMOV'S CODE OF ETHICS

1942:

- **First law:** Forbids robots from actively harming humans or allowing harm to come to humans by act refusal
- **Second law:** Obey humans unless they are in obstruction to the first law
- **Third law:** Protect themselves in accordance to first two laws

PICTURE FROM: [HTTPS://DE.WIKIPEDIA.ORG/WIKI/ROBOTERGESETZE](https://de.wikipedia.org/wiki/RoboterGesetze)



ASILOMAR AI PRINCIPLES (2017)

Formed
2017

Research

Ethics and
Values

Longer-
Term Issues

ASILOMAR AI PRINCIPLES (2017)

Research:

- Research Goal
- Research Funding
- Science-Policy Link
- Research culture
- Race avoidance

Ethics and Values (12):

- Safety
- Failure transparency
- Responsibility
- Personal Privacy
- Libery and Privacy
- ...

Longer-Term Issues:

- Capability Causion
- Importance
- Risks
- Recursive Self-Improvement
- Common Good

ETHICAL CHALLENGES OF AI

Technological
singularity

AI impact on
jobs

Privacy

ETHICAL CHALLENGES OF AI

Explainability

Responsibility

Fairness (Bias
and
discrimination)

Misuse

KEY AREAS FOR AI CODE OF ETHICS



Policy



Education



Technology

WAYS TO INCORPORATE ETHICS IN THE DEVELOPMENT?

How can you assure
that the use-case is
ethical?

What do you have
to keep in mind
during development?

How can you assure
that the use-case
stays ethical?

VALUE-BASED ENGINEERING (VBE)

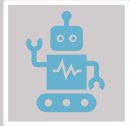
Business:

- Short-term shareholder value with profit maximization as core goals for management & developers
- Digital economy: primary services free (Facebook etc.), income through secondary sources (sharing of customer data)
- Secondary income often generates negative value externalities like loss of privacy, dignity and freedom
- If the product is only for making money, where is the purpose of life?

VBE:

- can help anticipate value harms that undermine success of innovation projects
- seeks to prohibit negative externalities
- supports identifying higher human and social values fostering attractiveness of primary value creation
- good requirement engineering very important before starting agile development

VALUE-BASED ENGINEERING: 10 PRINCIPLES



Principle 1: Ecosystem Responsibility

Cooperation with AI-provider on Algorithm design, selection process for training data, data quality assurance, AI's evolution and transparency



Principle 2: Willingness to Renounce Investment

Businesses should not be uneasy about their practices



Principle 3: Stakeholder Inclusiveness

Value reflection to help understand potentially adverse impacts

VALUE-BASED ENGINEERING: 10 PRINCIPLES



Principle 4: Use Moral Philosophies for Value Elicitation

Utilitarianism, Virtue Ethics and Duty Ethics



Principle 5: Context Sensitivity

Understand the systems context and anticipate its effects



Principle 6: Respect for regional laws and international agreements

Don't prioritize own continent's system values over and above their local customers expectations

VALUE-BASED ENGINEERING: 10 PRINCIPLES



Principle 7: Leadership Engagement

Transcending from profit-oriented thinking to a “healthy” profit



Principle 8: Transparency of the value mission

From core value principles to system requirements documented with a value register and an ethical policy statement summarizing the prioritized core values



Principle 9: Understanding Values in Depth

Conceptual analysis of prioritized core values for a better understanding

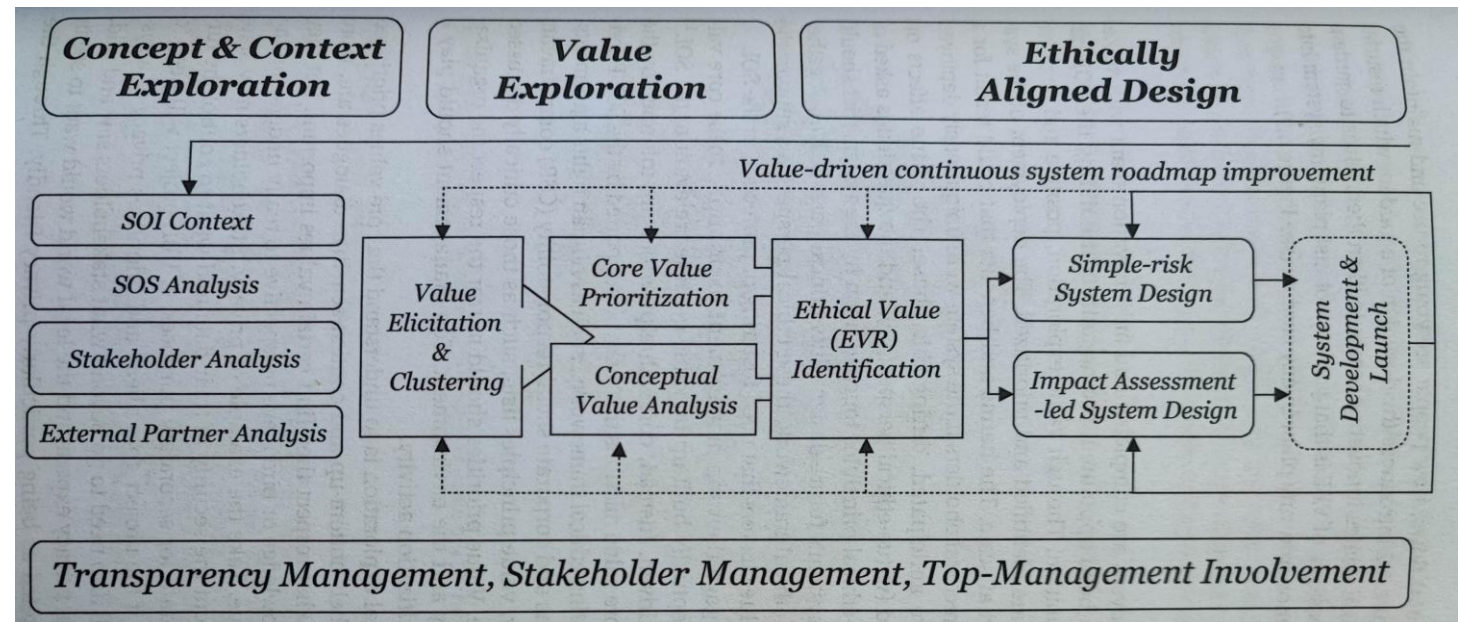


Principle 10: Using Risk-Analysis for System Requirement Elicitation

Prioritized core values to “Ethical Value Requirements” (e.g. qualitative descriptions) and translated into system requirements

VBE PROCESS OVERVIEW

- Three interlinked Blocks :
 - Concept & Context exploration
 - Ethical exploration and value prioritization
 - Activities translating core values into practical system of interest



IBM'S AI ETHICS



Principles:

The purpose of AI is to augment human intelligence

Data and insights belong to their creator.

AI systems must be transparent and explainable.



Pillars:

Explainability

Fairness

Robustness

Transparency

Privacy

WHAT ARE THE WAYS TO ACHIEVE ETHICAL AI?



LAWS



STANDARDS



CRITICAL THINKING
OF THE DEVELOPERS

AI DOOMERS

- Please read at home:
- <https://www.palladiummag.com/2025/01/31/the-failed-strategy-of-artificial-intelligence-doomers/>

RECAP

- What are some of the ethical challenges for AI?
- What is the core idea of value-based engineering?
- What are the ten principles of value-based engineering?

